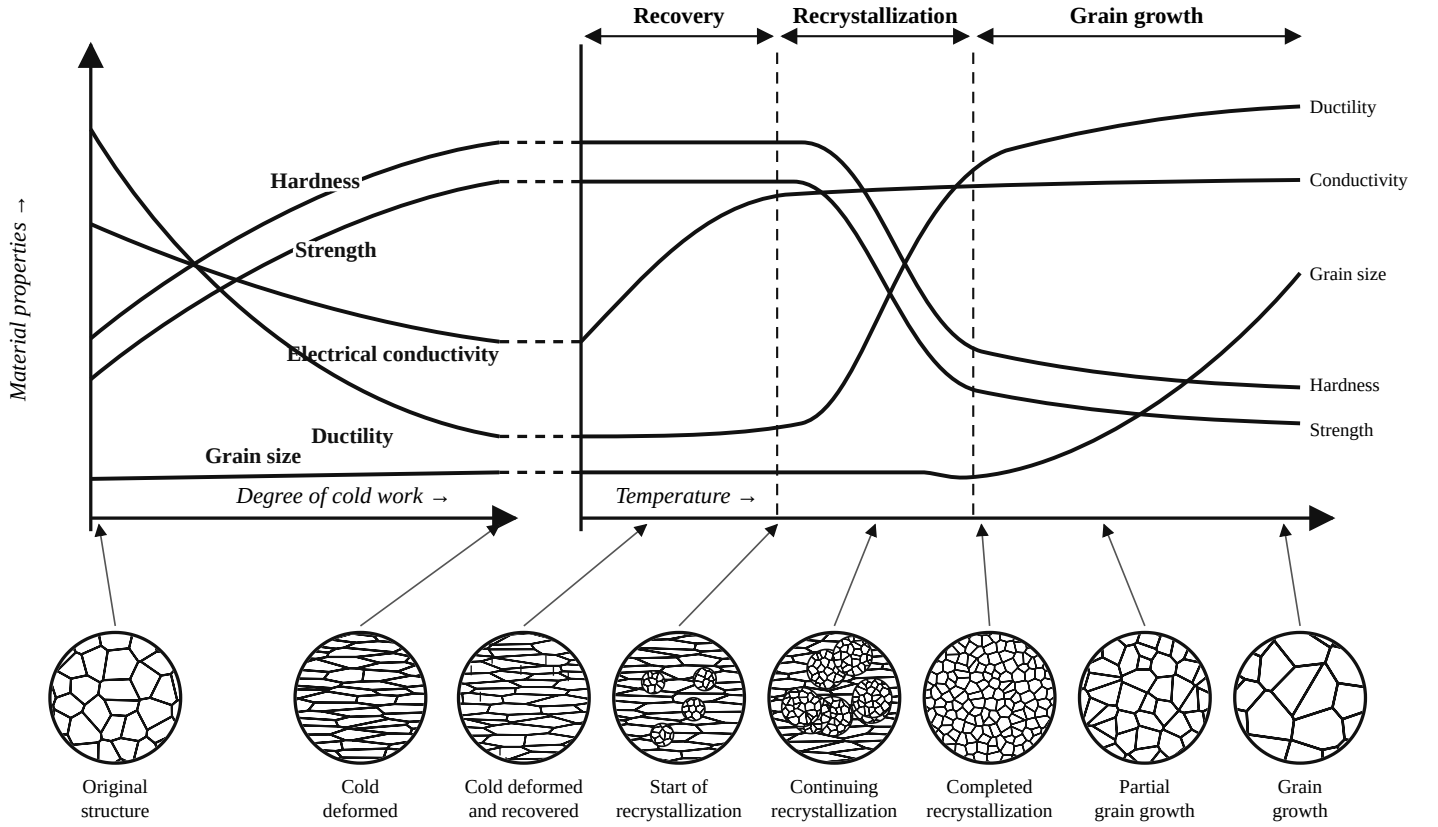
Exam Memorize Sheet

Q1 · Effect of Cold Work and Annealing on Material Properties



Effect of cold work and annealing on material properties

The diagram shows how a metal's properties change during **cold working** and then during **heating (annealing)**. As cold work increases, the grains become **elongated** and dislocations build up, so **hardness and strength increase** (strain hardening) while **ductility and electrical conductivity decrease**. On annealing, three stages occur: **(1) Recovery** – internal stresses are relieved and conductivity rises, but hardness, strength and grain shape barely change; **(2) Recrystallization** – new strain-free equiaxed grains replace the deformed ones, so hardness and strength drop sharply and ductility is restored; **(3) Grain growth** – at higher temperatures the new grains grow larger, further lowering strength and raising ductility.

Q2 · Periodic Table – Groups

	1A	2A	3B	4B	5B	6B	7B	8B	8B	8B	1B	2B	3A	4A	5A	6A	7A	8A
1	1 H Hydrogen																	2 He Helium
2	3 Li Lithium	4 Be Beryllium											5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
3	11 Na Sodium	12 Mg Magnesium											13 Al Aluminium	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon
4	19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
5	37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
6	55 Cs Caesium	56 Ba Barium	57–71 * Lanthanides	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
7	87 Fr Francium	88 Ra Radium	89–103 * Actinides	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson

Lanthanides	57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
Actinides	89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

1A–2A & 3A–8A = A groups (main group) · 3B → 2B = B groups (transition metals) · dashed boxes = inner transition metals

In the periodic table, the **vertical columns are called groups**. They are divided into two types: **A groups** (main group elements) and **B groups** (transition metals).

Special names of the A groups (main group elements)

- **Group 1A:** Alkali metals (except hydrogen)
- **Group 2A:** Alkaline earth metals
- **Group 3A:** Earth metals (Boron group)
- **Group 4A:** Carbon group
- **Group 5A:** Nitrogen group
- **Group 6A:** Oxygen group (Chalcogens)
- **Group 7A:** Halogens
- **Group 8A:** Noble gases (Inert gases)

Special names of the B groups (transition metals)

- **Between 3B and 2B (10 columns):** called **transition metals**, and all of them are metals.
- **Lanthanides and Actinides** are located at the bottom of the table and are called **inner transition metals**.

Q3 · Types of Atomic Bonds

Three types

- **Ionic bond**
- **Covalent bond**
- **Metallic bond**

Metallic bond (brief)

A metallic bond forms between metal atoms that give up their valence electrons. These electrons become a **"sea" (cloud) of free electrons** shared by all the atoms, and the positive metal ions are held together by their attraction to this electron cloud. The bond is **non-directional**, which explains why metals are **good conductors of electricity and heat**, **ductile** and **malleable**, and have a shiny lustre.